

# A Smart IoT-Based Automatic Load Sharing Transformer

Md. Murad Mia (2201139)  
Md. Mehadi Hasan Emon (2201140)  
Mahin (2201141)

Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology



## Abstract

Uneven load distribution in transformers can lead to overloading, reduced efficiency, overheating, and possible equipment failure. This project presents a Smart IoT-Based Automatic Load Sharing Transformer that automatically transfers loads between transformers under overload conditions. An ESP32 microcontroller, together with voltage and current sensors, is used for continuous monitoring and control. IoT integration enables real-time remote monitoring through a mobile application, enhancing system reliability and operational efficiency. For prototype validation, a 12-0-12 center-tapped transformer and a full-wave bridge rectifier (FWBR) were employed to supply DC loads, providing a safe and cost-effective laboratory-scale implementation of the proposed system.

## Project Background

Distribution transformers often experience uneven load conditions due to varying consumer demand, which may result in overloading, excessive heating, reduced efficiency, and shortened equipment lifespan. Conventional load management methods typically rely on manual intervention and provide limited real-time monitoring capabilities. With the advancement of IoT technologies, intelligent monitoring and automated control have become feasible for power distribution systems. This project explores the development of a Smart IoT-Based Automatic Load Sharing Transformer to enhance reliability, improve operational efficiency, and minimize the risk of unexpected power interruptions.

## Objectives

- To design an automatic transformer load-sharing system.
- To transfer loads automatically using relays.
- To improve transformer reliability and continuity of power supply.

## Required Components

01. Center Tap Transformer
02. Full Wave Bridge Rectifier
03. Current Sensor - ACS 712
04. Voltage Sensor - ZMPT101b
05. Relay Module
06. ESP32
07. I2C Display
08. Loads

## Methodology

- Two transformers supply the load through a relay-based switching network controlled by an ESP32.
- Voltage and current sensors continuously acquire load parameters, which are processed by the ESP32 for system monitoring.
- A 12-0-12 center-tapped transformer with full-wave bridge rectifiers provides the required DC supply for the control circuitry and prototype loads.
- When the measured current exceeds the predefined threshold, the ESP32 actuates the relay to transfer the load to the auxiliary transformer while updating the IoT monitoring interface.

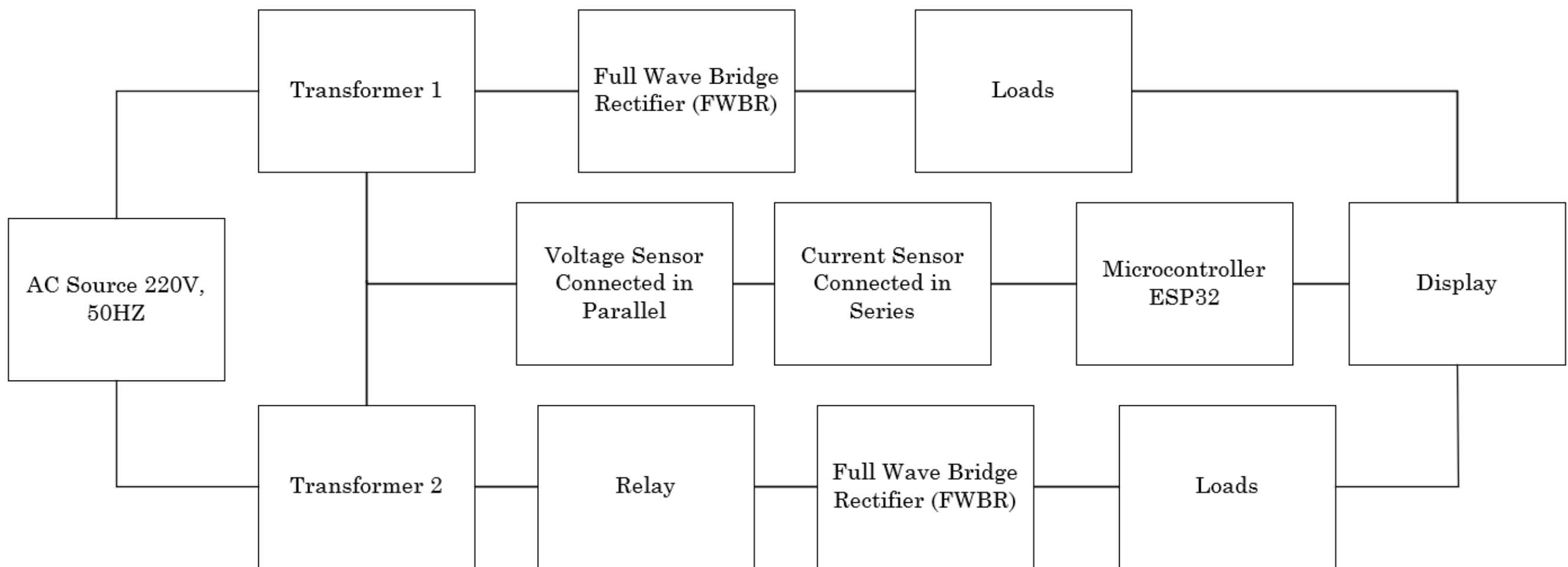


Figure 1: Block Diagram of the Project

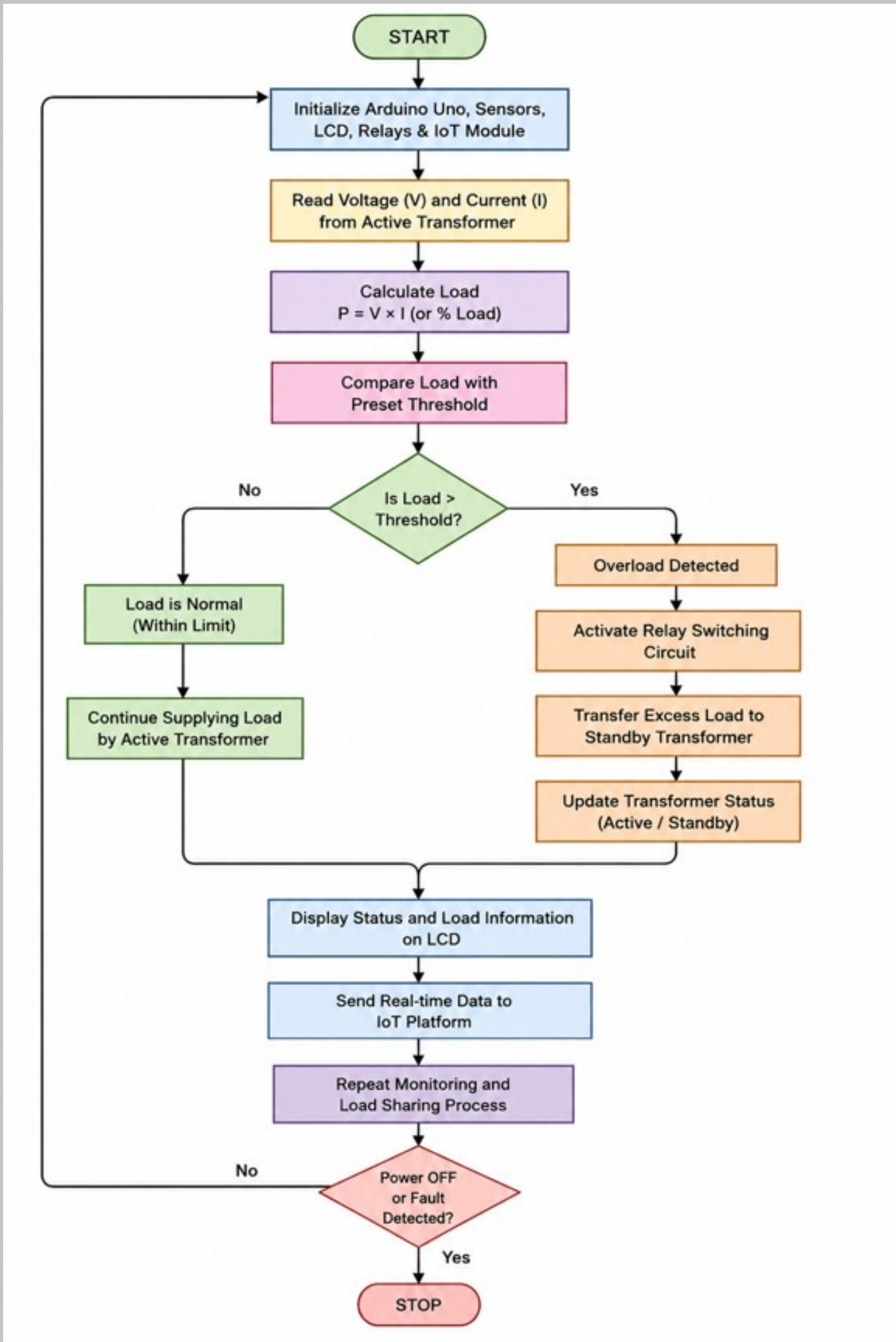


Figure 2: Flowchart for the complete operation

## Mathematical Interpretation

The system continuously monitors electrical parameters and performs decision-making based on threshold-based load sharing logic.

$$P = V \cdot I \quad (1)$$

where  $P$  is the real-time load power,  $V$  is measured voltage, and  $I$  is measured current from the sensors.

Transformer loading condition is evaluated using percentage loading:

$$L\% = \frac{I_{\text{load}}}{I_{\text{rated}}} \times 100 \quad (2)$$

Overload detection logic:

$$\text{Overload Flag} = \begin{cases} 1, & I_{\text{load}} > I_{\text{threshold}} \\ 0, & I_{\text{load}} \leq I_{\text{threshold}} \end{cases} \quad (3)$$

Relay switching decision for load sharing between Transformer-1 and Transformer-2:

$$R = \begin{cases} 1, & L\% > L_{\text{max}} \Rightarrow \text{Switch to Auxiliary Transformer} \\ 0, & L\% \leq L_{\text{max}} \Rightarrow \text{Normal Operation} \end{cases} \quad (4)$$

Sensor-based feedback model used by ESP32:

$$x(t) = \{V(t), I(t)\} \rightarrow \text{ESP32 Processing Unit} \rightarrow \text{Control Signal} \quad (5)$$

## Experimental Setup

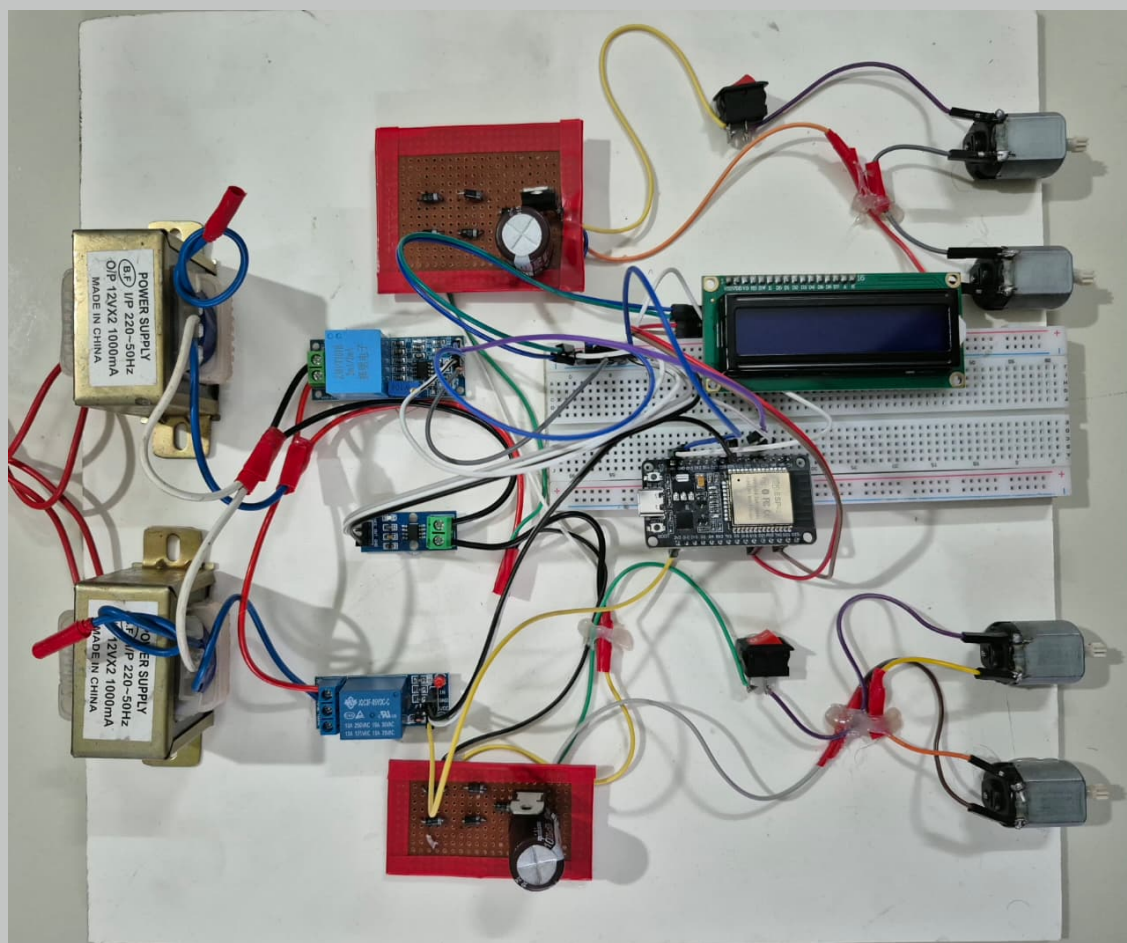


Figure 3: Practical Setup of the Project

## Results and Discussion

The proposed system was tested using two 12V DC outputs derived from separate FWBR rectifiers. Each rectifier feeds two DC motors connected in series as the load. Initially, both rectifier units are powered from Transformer-1 under normal operating conditions. The output power is calculated using:

$$P = V \cdot I \quad (6)$$

Under normal operation, the system maintains stable performance with both loads supplied from a single transformer. When the total load power exceeds the predefined threshold of 6.5 W, the ESP32 activates the relay switching mechanism, transferring the second FWBR load to Transformer-2. This ensures overload protection and continuous operation without voltage collapse.

### Sample Experimental Readings

| Condition          | Voltage (V) | Current (A) | Power (W) |
|--------------------|-------------|-------------|-----------|
| Normal Load        | 12.0        | 0.35        | 4.20      |
| Moderate Load      | 12.0        | 0.45        | 5.40      |
| Threshold Point    | 12.0        | 0.54        | 6.48      |
| Overload Condition | 12.0        | 0.60        | 7.20      |

### Threshold-Based Switching Logic:

$$\text{Relay State} = \begin{cases} 0, & P \leq 6.5W \quad (\text{Transformer-1 supplies both FWBR loads}) \\ 1, & P > 6.5W \quad (\text{FWBR-2 shifted to Transformer-2}) \end{cases} \quad (7)$$

The results confirm that the system responds effectively to overload conditions by redistributing load automatically and relay switching is stable.

## Conclusion

This project implements an ESP32-based IoT load sharing system that monitors voltage and current in real time and performs automatic relay switching during overload conditions. The system ensures uninterrupted operation through seamless load transfer between transformers. IoT integration enables remote monitoring, while the prototype using a 12-0-12 transformer and FWBR confirms a safe and cost-effective solution for intelligent load management.

## Future Scope

- Integration of advanced smart grid algorithms for adaptive load management.
- Replacement of relay switching with solid-state devices for faster response.
- Expansion to multi-transformer systems with cloud-based analytics and AI prediction.

## References

- M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4<sup>th</sup> edition.
- ESP32 Datasheet, Espressif Systems.
- IoT-based energy monitoring and smart grid research papers (IEEE Transactions on Power Systems).